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OCTOBER 15 1918

SERIAL NO. 165

THE MENTOR

RECLAIMING
THE DESERT

By C. J. BLANCHARD
of the
United States Reclamation Service

DEPARTMENT OF
SCIENCE

VOLUME 6
NUMBER 17

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THE MENTOR 1918.10.15, No. 165, Reclaiming the Desert

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Land and the Home-Coming Soldier

To the great number of returning soldiers, land will offer the great and fundamental opportunity. The experience of wars points out the lesson that our service men, because of army life, with its openness and activity, will largely seek out-of-door vocations and occupations. This fact is accepted by the allied European nations. That is why their programs and policies of re-locating and readjustment emphasize the opportunities on the land for the returning soldier. The question then is, "What land can be made available for farm homes for our soldiers?"



We have millions of acres of undeveloped lands that can be made available for our home-coming soldiers. We have arid lands in the West; cut-over lands in the Northwest, the Lake States and the South; and also swamp lands in the Middle West and South, which can be made available through proper development. Much of this land can be made suitable for farm homes if properly handled. But it will require that each type of land be dealt with in its own particular fashion. The arid land will require water; the cut-over land will require clearing, and the swamp land must be drained. Without any of these aids, they remain largely "No Man's Land." The solution of these problems is no new thing. In the admirable achievement of the Reclamation Service in reclamation and drainage, we have abundant proof of what can be done.



Our thought should now be given to the problem. We should know by the time the war ends, not merely how much arid land can be irrigated, nor how much swamp land reclaimed, nor where the grazing land is and how many cattle it will support, nor how much cut-over land can be cleared, but we should know with definiteness where it is practicable to begin new

irrigation projects, what the character of the land is, what the nature of the improvements needed will be, and what the cost will be. We should know also, not in a general way, but with particularity, what definite areas of swamp land can be reclaimed, how they can be drained, what the cost of drainage will be, what crops they will raise. We should have in mind specific areas of grazing lands, with a knowledge of the cattle that are best adapted to them, and the practicability of supporting a family upon them. So, too, with our cut-over lands. We should know what it would cost to pull or "blow-out" stumps and to put the lands into condition for farm homes. We should know what it will cost to buy these lands if they are in private hands. In short, at the conclusion of the war, the United States should be able to say to its returned soldiers: "If you wish to go upon a farm, here are a variety of farms of which you may take your pick, which the Government has prepared against the time of your returning."

From a letter to President Wilson from Secretary Lane of the Department of the Interior.

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A MIRACLE OF IRRIGATION—UNCOMPAHGRE VALLEY, COLORADO

RECLAIMING THE DESERT

Irrigation

ONE



rrigation, the artificial application of water to produce crops, is as old as agriculture. Genesis 1,17: "A river went out of Eden to water the garden." The practice of irrigation is probably coincident with that of the earliest agriculture of record for the reason that the latter was begun in regions of deficient rainfall in the Old World—Egypt, Assyria, Mesopotamia, Persia, India, Ceylon and China. Evidences abound of the existence of important storage and distribution systems constructed previous to 2000 B. C. Even greater antiquity is ascribed to similar work, the remains of which are found in the valley of the Euphrates, and also in China. The ancient aqueducts and subterranean canals of South America, extending for thousands of miles, once supplied great cities and irrigated immense areas. Irrigation on the Continent of North America was old when Rome was in the glory of its youth. Centuries before the venturesome Norseman landed upon the bleak and inhospitable shores of New England a large population dwelt in the hot valleys of the Southwest, in New Mexico and Arizona. From the solid rock, with primitive tools of stone, they cut ditches and hewed the blocks for many chambered palaces, which they erected in the desert or on the limestone ledge of deep river canyons. These voiceless ruins, older than the memory of many centuries, tell the story of a thrifty home-loving and semi-cultured people, concerning whose fate history brings us no word. The long lines of their canals, now choked with the wind-swept drift of centuries, give mute and pathetic evidence of the patience and engineering skill of the builders.

Early in the sixteenth century, when Coronado, the first great American explorer, swept up the Rio Grande Valley in Texas and New Mexico, he found a pastoral race dwelling in pueblos and practicing irrigation as had their forefathers, perhaps as far back as in the days of Abraham.

The priests with the aid of their Indian converts extended the practice of irrigation wherever they established the missions, and introduced many varieties

of fruits and vegetables from Spain and Italy.

Irrigation by English-speaking people began in America in 1700, and, for fully a century, was confined to use in the cultivation of rice on narrow strips along the coastal rivers of the Carolinas and Georgia.

The first Americans to reclaim extensive areas in the arid region were the Mormons, who, in 1847, settled on the eastern borders of the Great Interior Basin, on the site of Salt Lake City. Under wise leadership and through community effort, numerous canal systems were constructed and extended to thousands of acres of desert lands.

Between 1862 and 1880 many canal systems were constructed in Colorado, though none was important. The success of the Greeley colony, established in 1870 in the northern part of the State, gave considerable impetus to irrigation all over the arid region.

In the last thirty years there has been an awakening to the opportunity that lies in the arid West for the homemaker, and a remarkable transformation has taken place in many parts of this region. Irrigation canals long enough to girdle the globe three times now distribute the normal flow and the stored water of Western streams to millions of acres, which support hundreds of thousands of contented farmers. Cities, populous and great, have sprung up, rural communities, attractive and prosperous, broad vistas of fertile fields and blossoming orchards have replaced the wastes of sand and sage brush.

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HIGH LINE CANAL, UTAH VALLEY, UTAH

RECLAIMING THE DESERT

Skyline Canals

TWO



Locating and excavating several of the main canals proved almost as difficult and trying as the building of the big dams. This was especially true in the Yakima and Okanogan valleys in Washington, the St. Mary and Flathead in Montana, Uncompahgre and Grand River in Colorado, and the Utah valley, Utah. These canals cling to the edges of deep and precipitous canyons, or hug the steep mountain slopes. Here and there huge aqueducts of reinforced concrete span the chasms. For miles the water is siphoned across broad ravines in iron-banded stave pipe, or flows in tunnels through the cliffs.

The Tieton main canal of the Yakima project, Washington, for a distance of twelve miles is through a very rough country. It is cement lined throughout, and for two miles is in tunnels. For the greater part of its length it hugs the side of the canyon, in places 500 feet above the river. The lining for the canal was molded on the river bank, where water, sand, and gravel were available for concrete making, and the forms were carried to the top of the canyon by tram and cableways. The capacity of the canal is 300 cubic feet per second, and it is now irrigating 40,000 acres. The High line Canal of the Strawberry Valley project swings around the steep slopes of the Spanish Peaks of the Wasatch Range in Utah for several miles. At one point a portion of the water is dropped in pipes to the turbines, where power is developed and distributed to several towns in the valley. At various points along its course the canal is covered with a reinforced concrete roof to prevent injury and filling up from avalanches in the spring.

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GUNNISON TUNNEL. COLORADO

RECLAIMING THE DESERT

The Gunnison Tunnel

THREE



otable among the engineering works calling for courage and daring is the Gunnison Tunnel. This great bore, which in a sense changes physical geography, is excavated under the Vernal Mesa in western Colorado. The river portal is in the famous Black Canyon of the Gunnison, a precipitous chasm 3,000 feet deep, and the tunnel extends for nearly six miles, 2,500 feet below the surface of the Mesa. The outlet is in the valley of the Uncompahgre. By means of a low diversion dam, the waters of the Gunnison are turned into the tunnel and transferred to the desert valley of another stream. The early reconnaissance surveys and examinations and the final work of determining the location of the tunnel were attended with unusual dangers. The exploration of the canyon, which up to that time had never been traversed by man, and the careful and detailed preparation of a topographic map of this rugged region called for genuine heroism. In the former task the surveyors risked their lives for many days in the depths of the gorge, and, in the latter, the engineers performed their duties under conditions of great hazard and peril.

The construction forces met and overcame almost every difficulty ever encountered in tunnel excavation. Gas, cave-ins and subterranean springs interposed obstacles throughout the work. At one time a heavy flow of carbon dioxide, or choke damp, forced the workmen to flee for their lives and delayed operations until a ventilating shaft 680 feet deep was sunk. The heavy flows of hot and cold water necessitated the use of large pumps for months at a time. For more than 500 feet the tunnel was driven through a remarkable bed of fossils, consisting of the shells of extinct sea creatures, many of which were of great size. Exposed to the air, disintegration was rapid, and the huge masses of falling rock imperilled the lives of the workmen.

Notwithstanding the extraordinary care of the Government to safeguard the employees, a heavy toll of human lives was taken. The excavation work was projected from four adits, three on the Uncompahgre side and one in the

canyon, and crews of drill men and assistants simultaneously began boring into the mountain from both sides. When the last charge of dynamite had been exploded and the tunnel was “holed” through, it was found that the line was true within a fraction of an inch.

Connecting with the outlet of the tunnel is a large cement-lined canal which conveys the water to the valley ten miles below. Here it mingles with the Uncompahgre, and by means of a complex distribution system of canals and laterals, finally reaches the irrigable lands.

The Uncompahgre Valley is one of exceptional scenic beauty, and is blessed with a fertile soil adapted to the growing of a wide variety of products. Its irrigable area is approximately 100,000 acres, more than half of which is now producing two generous harvests.

The discharge of the Uncompahgre River was quite inadequate for the irrigable lands in the valley, which had been brought under the ditches constructed before the passage of the reclamation act, and failure of crops occurred frequently during the low water periods. In addition there were thousands of acres of desert land in the valley doomed to aridity unless the water supply could be supplemented. The Gunnison River, flowing uselessly in its profound canyon on the other side of the range, was drawn upon by means of the tunnel, and an abundance of water is now assured for all the lands included in this project. Agriculture here is exceedingly diversified, and in many sections, is intensive. The products of the soil range include alfalfa, cereals, sugar beets, potatoes, onions, peas, beans, and other vegetables, peaches, pears, and apples. Surrounding the valley are vast areas of fine grazing lands included in national forests, which furnish summer pastures for thousands of cattle and sheep. The rapid settlement of the farm lands has been followed by a corresponding growth of the towns of Montrose, Delta and Olathe.

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ROOSEVELT DAM, ARIZONA

RECLAIMING THE DESERT

The Roosevelt Dam

FOUR



First among the imposing structures of the Service in the order of beginning and completion is the Roosevelt Dam, the important engineering feature of the Salt River project in Arizona. Brilliant and daring in its conception, diverse and complicated in its problems, and stupendous in its structural features, the Roosevelt Dam stands today an enduring monument to the creative genius of its designer and builder. The layman will not appreciate the complexity of the problems and the variety of obstacles that were encountered without an understanding of the locality in which the work was carried on. The dam is located in a region which was once regarded as almost inaccessible except to the nomadic Apache, who found a safe refuge here for many years. The nearest railway is 62 miles away. Twenty miles of this distance is across a waterless, parched and forbidding desert. For more than 40 miles the country, gashed and fissured into fantastic forms, a region of stupendous canyons, steep sided mountains, and turbulent streams, presented an almost insuperable barrier to ingress. It is a region of colorful and inspiring scenery, but most unpromising as a site for a large engineering work. The initial step was to build a highway to the dam-site, a task of much difficulty on account of the rough country and the unwillingness of white laborers to do the work. The road was finally completed, largely by utilizing the Apache Indians, many of whom were remnants of Geronimo's band of marauders. The influence of this experiment in employing Indians has been of lasting benefit to the Red Men of the Southwest, many of whom have continued on similar work for railroads and other corporations. The Roosevelt Road, now familiarly known as the Apache trail, is one of the most spectacular highways in the country, and thousands of tourists go over it each year in automobiles.

The site for the dam was located at the entrance of the profound canyon that Salt River has cut through the mountains. Just above the canyon, Salt River and Tonto Creek converge in a broad, level valley, which is now submerged to a

depth of nearly 200 feet, forming one of the largest artificial lakes in the country.

The Roosevelt Dam is of sandstone hewn from the walls of the canyon in which it is built. It is of gravity section, arched up stream, 280 feet high, 168 feet thick at base, and 1,080 feet long on top. At its base it covers an acre of ground. A building 209 feet square and 26 stories high would about cover the space of the dam except that halfway up the sides there would be space for two more structures, each 11 stories high and 885 feet long on top. Owing to the long haul for the railroad the activities of the engineers were varied and numerous. The Government cement mill turned out 600,000 barrels of first-class cement, the saw-mill in the National Forest manufactured all the lumber required to house 2,000 people, and for stores, offices, etc.; a hydro-electric plant furnished power to the contractor on the dam and light for the camp; two farms were operated to supply food for the employés; water works and sewerage systems were installed, and law and order were preserved during four years of construction. The engineer had charge also of a large commissary, a big mess, and superintended a brickyard and cement pipe plant.

During the building of the dam the valley below was the scene of unprecedented activity. A million-dollar diversion dam was constructed across the Salt River to divert the stored water into thousands of miles of canals; power plants, pumping plants, transmission lines, and a thousand and one engineering details were completed in advance of the great day when the turbulent floods of Salt River would be conserved and led to the thirsty lands. On March 18, 1911, former President Roosevelt, in the presence of an assemblage of nearly 1,000 people, formally dedicated the structure which fittingly bears his name. By the simple pressure of an electric button the enormous gates weighing 60,000 pounds were raised and released the pent-up floods for irrigating nearly 200,000 acres of Salt River Valley. The swift passing of years has been marked by marvelous progress in this desert valley. In 1902, when the work was started, the assessed valuation of the country, of which the valley is the larger part, was \$5,000,000. In 1916, the taxable property values were \$72,000,000. In 1913, the first crop census of the project was taken and showed an irrigated acreage of 159,170, and a gross value of crops of \$4,775,000. In 1917, the total acreage watered was 201,600; the gross value of crops was \$13,692,000. During the same period the number of farms increased from 3,600 to 4,326. The net cost of the entire project to June 30, 1917, including \$3,500,000 for the Roosevelt Dam, was \$11,367,000. The annual returns from the land irrigated by it are more than \$3,000,000 in excess of this amount. The gross value of crops in

1917 was almost equal to that of New Hampshire and Rhode Island in the census year of 1909. In October of 1917 the Roosevelt Dam, canal system, and power plants were formally transferred to the Water Users' Association, under whose management the project henceforth will be operated.

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ARROWROCK DAM, IDAHO—THE HIGHEST DAM IN THE WORLD

RECLAIMING THE DESERT

The Highest Dam in the World

FIVE



Arrowrock dam, in Idaho, is probably the most spectacular structure to the credit of the Service. Completed on October 4, 1915, it ranks all other dams in the world in its height, 350 feet above bedrock. It is of rubble concrete, arch gravity type, and contains 585,130 cubic yards of material. It was built by Government forces, and not by contract, and its completion in two years less than the estimated time and at a saving of more than two million dollars in the estimated cost, furnishes a striking example of Federal efficiency and economy. In connection with this important work the engineers built and operated a standard gauge railway 14 miles long, which carried more than 80,000 passengers, and 14,000,000 tons of freight. A unique camp, containing 4,000 people, was established, with sewerage, water works, and electric lights. Schools for the children of employees, a hospital, postal savings bank, churches, and Y. M. C. A., a large general store and commissary, blacksmith and machine shops, sand-grinding and cement-mixing plants, all under the engineer's direction, gave to the camp the aspect of an enterprising and busy community. It was a camp in which, for four years, there was no night. Throughout the greater part of this period the work proceeded without interruption, night and day, with three eight-hour shifts. Profiting by the experience gained on other large works of the Service, and with labor-saving devices of their own invention, the engineers on Arrowrock worked with extraordinary swiftness and sureness, and established a most enviable record for economy and time. The total cost of the Arrowrock is approximately \$5,000,000, and its principal purpose is to conserve the floods of Boise River for the irrigation of 240,000 acres of land embraced in the Boise project.

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SUBURB OF PHOENIX, ARIZONA—RECLAIMED FROM DESERT LAND

RECLAIMING THE DESERT

The Romance of the Desert

SIX



vein of romance runs through every form of human endeavor. No life so sordid, prosaic, or wretched, but has felt sometime its light and gladsome touch. In the desert, romance finds its chief essentials in adventure, courage, daring and self-sacrifice. For more than half a century man has been writing a romance of compelling interest upon the face of the dusty earth. Irrigation, with Midas' touch, has changed the desert's frown to smiling vistas of verdure. Its solemn silence has been broken by the voices of countless happy people.

Our national strength is in its citizens, and the place in which the best is bred is in the country. The threat of urban congestion, no longer remote, and the pressure of population, which even now is bearing heavily upon our resources, are unanswerable arguments for increasing and making permanent the nation's virility, prosperity and growth by creating more country homes.

It is an economic axiom that the stability of a nation is assured only when the bulk of its citizens reside in their own homes. The ideals and principles for which our forefathers fought cannot be preserved and maintained by a citizenry whose interest does not extend beyond mere wage earning. The American desert was won by war, treaty, discovery and purchase. Flying at one time the flags of four nations, its history is rich in thrilling incident and adventure. Its milestones are the bones of trappers, explorers, and pioneers. Its people are strong and courageous. To battle with the elemental forces of nature has become a passion. While the glamor of romance in years ago is dispelled, it is still romance-land, but with a new background. The romance of creation now pervades the once silent desert, and the dominant thought and impulse is to establish there the well ordered life of New

England with all the highly organized facilities for making existence in the country attractive.

American people cannot rightly claim to have measured up to their opportunity until the deserts of the West and the swamp lands of the South have been replaced by vistas of prosperous farmsteads.

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SERIAL NUMBER 165**

RECLAIMING THE DESERT

By C. J. BLANCHARD

United States Reclamation Service

With Illustrations from Photographs Supplied by the United States
Reclamation Service.

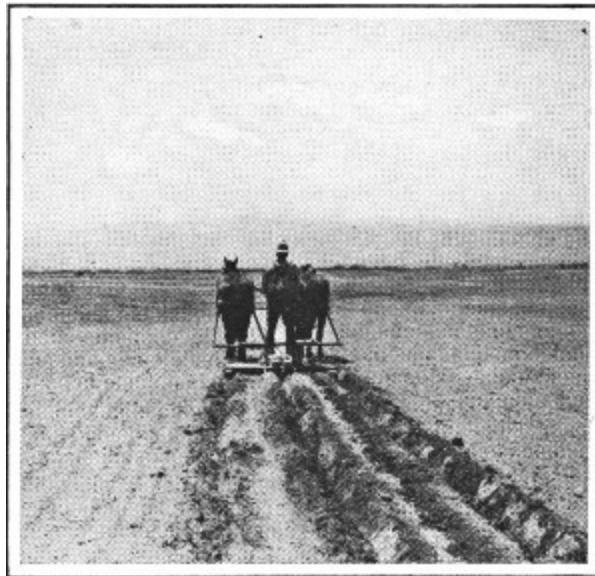
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MENTOR GRAVURES

A MIRACLE OF
IRRIGATION,
UNCOMPAHGRE
VALLEY, COLORADO
HIGH LINE CANAL,
UTAH VALLEY, UTAH
GUNNISON TUNNEL,
COLORADO

MENTOR GRAVURES

ROOSEVELT DAM,
ARIZONA
ARROWROCK DAM,
IDAHO
SUBURB OF
PHOENIX, ARIZONA



MAKING FURROWS FOR IRRIGATION

"It is a grander achievement to expand the domain of civilization by water than by blood."



ational reclamation was the dream of Western statesmen and thinkers for a quarter of a century before a laggard Congress gave it form and actuality by the law of June 17, 1902. With the passage of that law and another which initiated the construction of the Panama Canal,—both were signed by President Roosevelt in the same month,—the engineering forces of the nation were flung into widely differing fields of activity. With the Panama Canal engineers, the task, though herculean, was confined to a restricted and perfectly well defined area. On the other hand, the Reclamation problems were generally in regions widely separated, remote from transportation, and often unsurveyed and unexplored. To appreciate the variety and magnitude of the tasks involved, it is necessary briefly to describe the general character of the country in which these works were projected.

The Desert States

The great American desert may be roughly described as lying between the western boundary of the Mississippi Valley and the Pacific Ocean, and as embracing about two-fifths of the total area of continental United States, exclusive of Alaska. The superficial area of the several states that comprise this desert is almost equal to the combined areas of Norway, Sweden, Denmark, Holland, Germany, Belgium, France, Italy, the British Isles, Austro-Hungary, Spain, Portugal, the Balkan States, Turkey in Europe, and Japan. The population of the so-called desert states^[A] is 16,423,625, while that of the countries above-mentioned is over twelve times as great. Within the confines of the desert is every gradation of climate from north temperate to semi-tropic found in these European countries. Its physical geography includes a wide variety of features from the Great Plains to the highest and lowest elevations in our country. Herein are found the most notable scenic attractions, including the great national playgrounds of Yellowstone, Glacier, Rocky Mountain, Mount Rainier, Crater Lake and Yosemite National Parks, the Grand Canyon of Arizona, and the principal national forests.

^[A] Arizona, New Mexico, California, Oregon, Washington, Nevada, Utah, Idaho, Montana, Colorado, Kansas, Nebraska, North and South Dakota, Oklahoma.



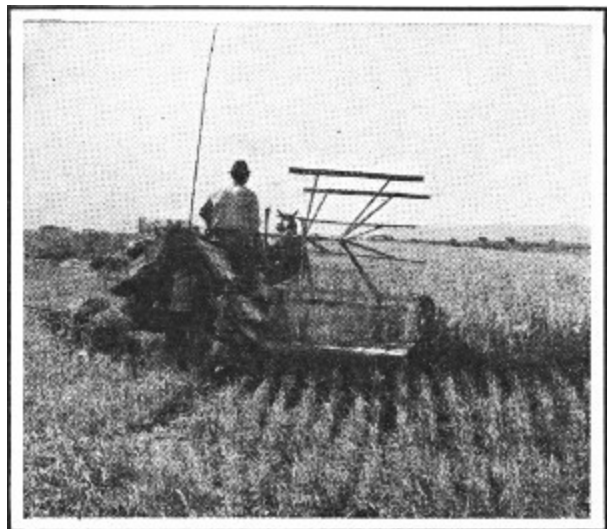
THE FIRST FLOW OF WATER

In Grand Valley Project, Colorado

As to water supply, the desert belongs in two regions, arid and semi-arid. West of the Missouri and extending to the foot of the Rocky Mountains, is a vast area of foothills and gently sloping, grassy prairies, which constitute a large part of the semi-arid or sub-humid region of the United States. This broad belt, stretching from Mexico northward into Canada, has no clear-cut boundary separating it from the humid region on the east or the arid region on the west,

owing to the variance of the mean annual precipitation in many localities. A convenient and easily marked line for the eastern boundary of the arid region is one closely following the hundred and third meridian. On the north it bends away from the meridian toward the west, and on the south tends eastward north of the Rio Grande. On the west the arid region extends to the Pacific Coast in extreme southern California, but from Monterey north there is a narrow belt of semi-arid and humid country bordering the Pacific Ocean. The Coast Range is humid, especially in Oregon and Washington, where the rainfall is as heavy as in any other part of the United States.

The true desert, wherein the production of crops is wholly dependent upon the artificial application of water, lies for the most part west of the Rocky Mountains. Its estimated area, 900,000 square miles, is probably slightly greater than that of the semi-arid region. An accurate determination of the relative size of each, however, is not possible until a comprehensive hydrographic survey has been completed.



Contrary to popular opinion, this region is not a vast wilderness, desolate and unpromising. Between the mountain ranges lie many beautiful valleys, through which flow numerous streams fed by melting snows. Utah and Nevada contain large areas of sage-brush desert, comprising what is known as the Great Basin, which has no outlet to the sea, and is doomed to aridity by reason of the absence of living streams. Millions of acres of desert now inhabited by the coyote and the rattlesnake await only the coming of the engineer to wake to teeming fecundity.

FORTY BUSHELS OF WHEAT PER ACRE

Shoshone Project, Wyoming. This was desert land in 1915

In one important respect the arid and semi-arid regions are alike, and that is in the character of the streams. Almost without exception the rivers of the West are erratic and “flashy” in flow, are subject to long periods of drought and sudden and destructive floods. The full utilization of these streams for irrigation and power necessitates storage. In the control of floods, engineers have found their greatest problem, and one whose solution has taxed the skill, imagination and daring of twentieth-century genius.



WHEAT ON SHOSHONE PROJECT,
WYOMING

This land was desert in 1908

in the first foot-note, with Texas added.

National Irrigation

National irrigation became a fixed policy of the American Government with the passage of the Reclamation Act in June, 1902. The principal provisions of this act are briefly as follows:

First. A reclamation fund in the Treasury consisting of the proceeds from the sales of public lands in the sixteen arid and semi-arid states. [\[A\]](#)

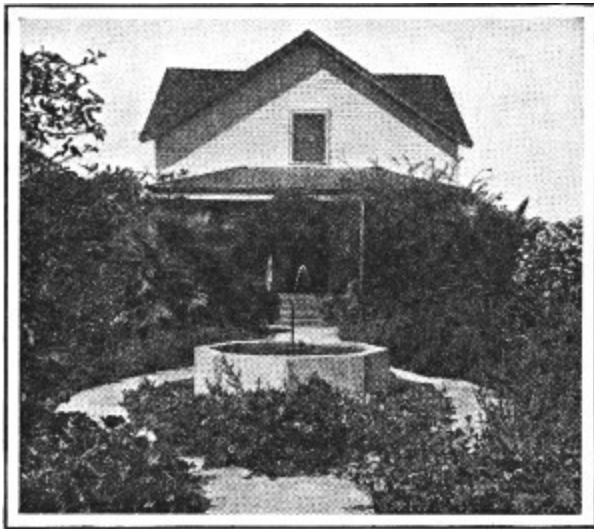
[\[A\]](#) These include the states named

Second. A Reclamation Service in the Department of the Interior to investigate and report on the irrigation projects to the Secretary of the

Interior, who, with the approval of Congress, may authorize construction and let contracts, providing the money is available in the fund.

Third. The return to the fund of the actual cost of each project by the sale of water rights, payments to be made in a series of instalments running over a period of twenty years without interest. The money so returned is to be used again and again on other works.

Fourth. The holding of public lands for actual settlers under the



A COMFORTABLE "DESERT" HOME

Having electric power for heat, light and cooking



SAGE BRUSH AND ALFALFA, MINIDOKA PROJECT, IDAHO

On one side of the fence is the desert sage brush, on the other rich alfalfa

homestead act in small farm units sufficient to support a family.

Fifth. The sale of water rights to private land owners, but not for more than 160 acres.

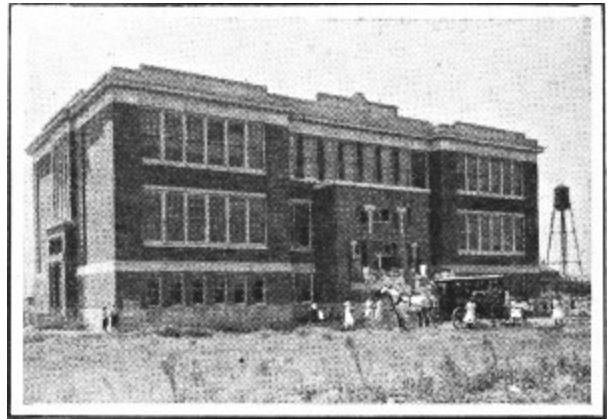
Sixth. The ultimate turning over to the people of the irrigation system, to be operated and managed by them under a system of home rule.

The policy of national irrigation is broadly paternal and of enormous economic importance to the whole country. In the building of new commonwealths in the arid West the Government is utilizing its own

undeveloped resources. It is creating opportunities for its citizens to establish themselves in permanent homes in which patriotism, loyalty and civic pride are bred and fostered. For a number of years the growth of population has been abnormal when compared with the development of the

agricultural industries that must support the people. Farming as a profession has been languishing and falling behind the general development of the country. The rapid increase in land values has made it correspondingly more difficult for the man of small means to acquire a foothold on the land. Practically every progressive nation in the world has come to recognize this fact, and is making provisions to encourage and assist its citizens to undertake farming. The primary purpose of the Reclamation Law, therefore, is to make homes on the

land. To the new empire in the West have flocked the young, the strong, the adventurous, and herein we are witnessing a gradual welding of all the Aryan races into a final race type. Signs are not lacking that this type in time will dominate the world, for the desert offers to every man his true birthright—room to breathe, sunshine, a sure reward for intelligent labor, the individual home, and an opportunity to become independent. Desert reclamation already has gone beyond the stage of prophecy. The material and substantial results that have been accomplished place the work of the Government on a practical and solid foundation.



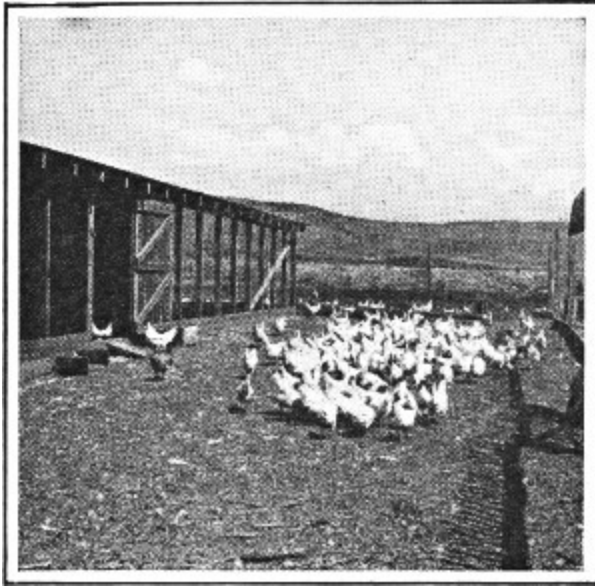
HIGH SCHOOL BUILDING, MINIDOKA
PROJECT

This beautiful, fully equipped school stands
where there was only desert in 1907

The Romance of Reclamation

The history of national reclamation is as interesting and romantic as a tale from the Arabian Nights. Romance colored the vision of builders that saw in the sparkling streamlets, the unchecked floods, the wide, free plains and the vacant mountain valleys a promise of independence, happy homes and laughing children. Theirs was not the incentive of large emoluments, for Government salaries are notoriously meager. Their inspiration came from doing a signal work of splendid usefulness,—conquering nature in her unfriendliest mood for the permanent and lasting good of mankind. As they toiled in the fastnesses of the mountains, in abysmal canyons or far out in

the voiceless desert, through the blazing summer heat of the Southwest or the fierce blizzards of the northern plains, this thought was uppermost, "By this work we shall make the desert blossom."



CHICKENS THRIVE

In the dry climate of the desert

a few of these interesting irrigation projects of the United States Federal Government are here mentioned, although each is worthy of extended description. The annual reports of the Service are obtainable from the Superintendent of Public Documents in Washington, and these contain full details.

In the order of their magnitude and the spectacular character of the engineering work, the Arrowrock dam in Idaho, Elephant Butte dam in New Mexico, Roosevelt dam in Arizona, Shoshone and Pathfinder

Their dramatic achievements stand out boldly in this age of engineering triumphs. The mighty floods of western rivers have been checked behind enormous masonry dams, several of which are ranked among the highest in the world. Physical geography has been altered by transferring rivers from one drainage basin into that of another. Whole rivers are now flowing through tunnels that pierce lofty mountain ranges, and the water is being distributed in thousands of miles of canals to a million acres of desert.

Owing to limitation of space only



FIRST SETTLERS ON DESERT LANDS

Settlers' houses where once was the haunt of the buffalo

dams in Wyoming, and Gunnison tunnel in Colorado, take first rank. Several of these are described at length elsewhere.



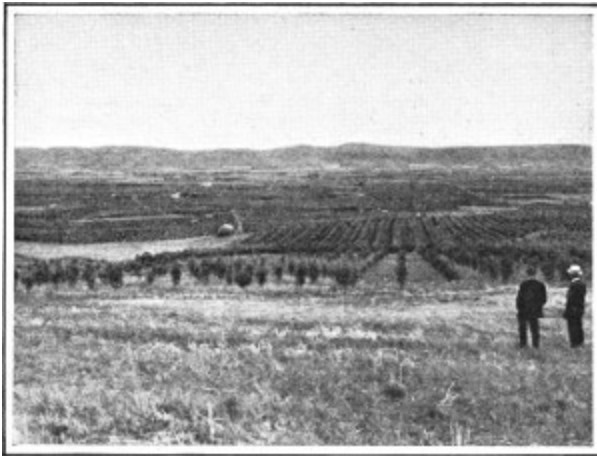
AN OREGON MELON PATCH

The Most Capacious Irrigation Reservoir

In a region rich in thrilling reminiscence along the pathway trod by the Conquistadores of Spain, Federal engineers recently completed a monumental structure of masonry known as the Elephant Butte dam in New Mexico. This enormous mass of rock and cement effectually and permanently blocks the canyon of the Rio Grande just below the isolated basaltic peak

from which it derives its name. It rises from the depths of the canyon 318 feet, and its crest length is 1674 feet. It owes its place among the greatest structures of the age to the enormous capacity of the reservoir created by it. Behind this towering monolith the greatest floods of the turbulent Rio Grande are held in a lake forty-five miles long and four miles wide. When full it will contain enough water to cover 2,627,700 acres to the depth of a foot, or sufficient to submerge the entire State of Connecticut ten inches deep. This stored water, when needed for irrigation, is turned back into the river and taken out at several points above and below El Paso, Texas; 180,000 acres in New Mexico, Texas and Old Mexico are being brought into cultivation. The charm of antiquity pervades the whole region. Here irrigation was practiced long before the first written word of our history. Centuries before the coming of the Spanish missionaries, a pastoral race dwelt here and cultivated this fertile valley along the stream. A later civilization merely absorbed and extended the primitive canals until the era of national reclamation aroused the valley to new life and purpose. While some of the primitive methods of agriculture, differing but slightly from those of Biblical days, are still practiced, modern harvesting machinery is replacing the hand sickle, the sulky plow supplants the sharpened stick, and the threshing of grain is now performed by modern methods, and rarely by

means of goats and ponies. Splendid highways of concrete and macadam connect the farming communities with numerous thriving towns, and the quaint groups of adobe houses, which here and there rise in the desert, are the last remains of vanishing races that are slowly giving way to modern progress.



SIX YEARS FROM DESERT LAND

A rich and fruitful farm on land barren in 1912

The Shoshone Dam

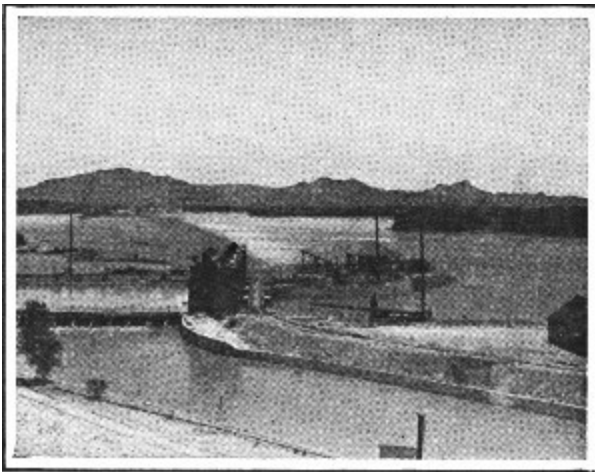
In northern Wyoming, near the eastern entrance to Yellowstone National Park, the South and North Forks of Shoshone River plunge downward from the steep slopes of the Rockies and unite in a broad level-floored valley which in early geological times was a beautiful mountain lake. At the lower end of the valley the river rushes abruptly into a deep and narrow canyon. The entrance to the canyon, which is only sixty feet wide at the bottom,

was selected as the site for the Shoshone dam. Before construction began it was necessary to blast a highway through the gorge for a distance of eight miles, in order to connect the work with the nearest railway. This highway is now a part of the road system into the Park, and is known as Cody Way.

Nearly a year was spent in investigating bed rock conditions before the foundation was placed. Enormous boulders eighty feet in thickness were discovered in the bed of the stream; the removal of these, owing to the narrowness and depth of the canyon, proved expensive and difficult. Before any work was undertaken in clearing the channel of obstructions, the river was passed around the dam site by means of a tunnel. During the early period of construction the laborers were taken to and from their work, hundreds of feet below the camp, by means of baskets, and skips were suspended on cable ways across the gorge. Sudden floods, extreme cold weather, short seasons, and other unfavorable conditions greatly retarded progress.

The dam is a wedge of concrete, with curve up stream, 328 feet in height, and 200 feet long on top. It has restored the lake of olden times, and this beautiful body of water is now one of the attractions of the trip to the Park. The stored water will be utilized for the irrigation of 150,000 acres of land on the mesa (a high, broad, flat table-land) below Cody. This region until 1907 was a worthless and uninhabited desert. Today it is occupied by more than 700 farm families and three growing towns.

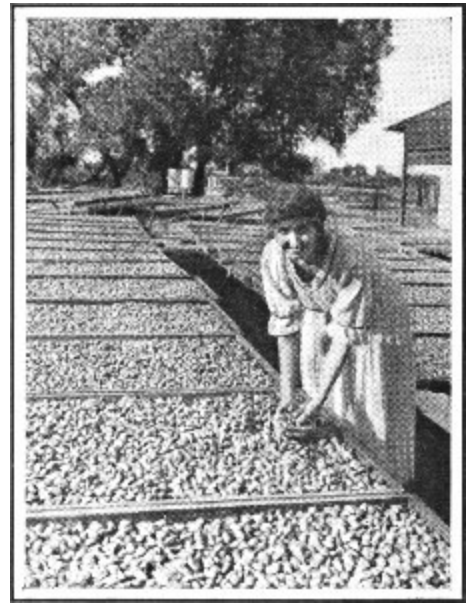
Conquering the American Nile



LAGUNA DAM, ARIZONA-CALIFORNIA

Length, 4781 feet; 19 feet high. Division Gates, 17×33 feet, are the largest in the world

is nearly a mile long, nineteen feet high, and 267 feet wide on the bottom. Its type is similar to that of the barrages of the English engineers in India. Laid on a foundation of shifting sand and silt, it is held in place by its enormous weight, 600,000 tons. Owing to the quantities of silt carried by this stream, the dam is provided with a large settling basin, sluiceway and



DRYING ALMONDS

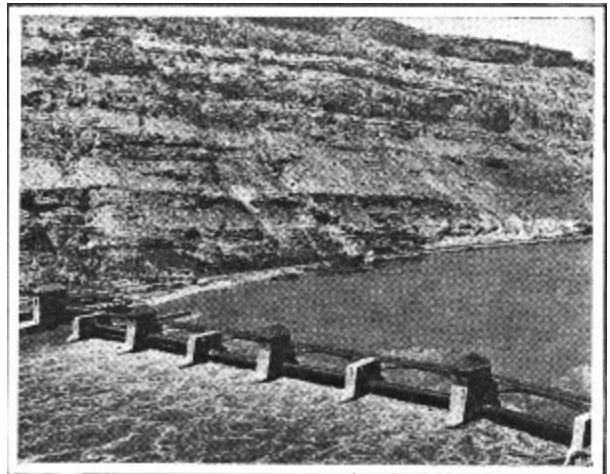
On the Orland Project, California

The delta of the Colorado River, in Arizona and California, is often described as America's Valley of the Nile. In climate, soil and agricultural conditions it is singularly like the great valley of Egypt. The conquest of this turbulent river of the West was a splendid achievement. The engineering features of the Yuma project, in Arizona and California, are in many respects original. The control of the stream called for a long diversion dam and more than 100 miles of strong levees. The dam

gates, by which comparatively clear water is turned into the canals. The main canal system extends for twelve miles on the California side. Just opposite the city of Yuma, Arizona, the entire volume, 1,000 cubic feet per second, is dropped in a siphon 1,000 feet long and carried under the river to the Arizona side. Here are 90,000 acres of valley and mesa land, and more than half of the area is being developed intensively. The soil is of great depth and extremely fertile, and the climate is adapted to the growing of a wide variety of crops. The mesa lands which are to be opened later are described as frostless, and especially adapted to growing citrus and other semi-tropic crops.

The Pathfinder Dam

Far from the beaten path of man, fifty miles from the nearest railway in a deep and narrow granite canyon of the North Platte River, in southern Wyoming, engineers in 1903 located an admirable site for a high masonry dam. Early maps of the expedition of Gen. John C. Frémont, the explorer, indicated the spot as the scene of a disaster where these adventurers suffered the wreck of their boats and the loss of many belongings in the rapids of the river. Hence the name Pathfinder dam. This is a beautiful structure of huge granite blocks 225 feet high and 600 feet long on top, and its cost was \$1,000,000.



GRAND RIVER, ROLLER-TOP DIVISION
DAM, COLORADO

Electrically operated rollers are lifted to pass
the floods

Judged from the service it has rendered mankind already, as a preventive against disastrous floods and a guarantor of generous harvests, it deserves a high place among the storage structures of the world. Since its completion the North Platte River has been completely tamed, and angry floods that once wrought millions of dollars of destruction are now so distributed as to



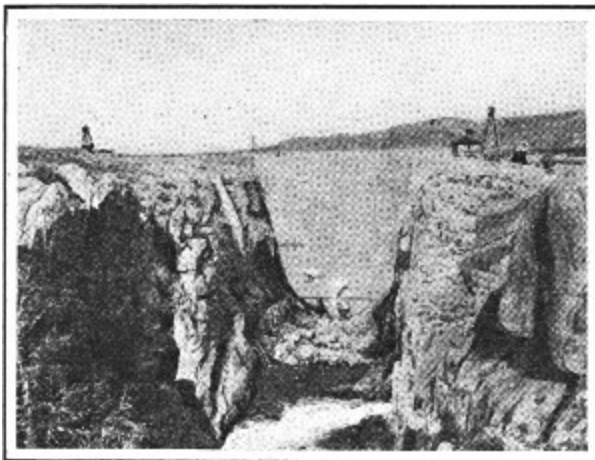
SHOSHONE RESERVOIR, WYOMING

insure an annual harvest valued at \$6,000,000 in a region once occupied only by nomadic herdsmen.

Transferring a River

The waters of the Strawberry River in Utah, which for ages flowed idly to the Pacific Ocean, are today contributing to the material prosperity of a beautiful valley in the drainage of Great Salt Lake. The transference of a river

from its own drainage into that of another has been performed by the Federal engineers on several occasions, notably in Colorado, where the Gunnison River is augmenting the flow of the Uncompahgre, the St. Mary River in Montana (a former tributary of the Arctic Ocean, now transferred to Atlantic drainage) and the Truckee River in Nevada, now consolidated with the Carson River.



PATHFINDER DAM, WYOMING

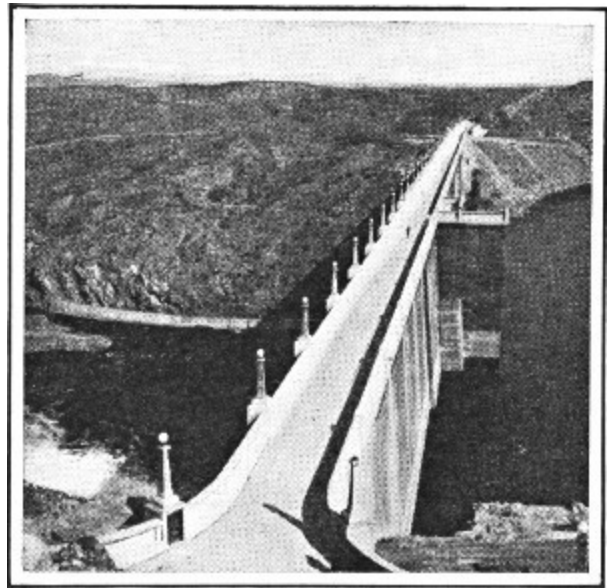
218 feet high, 600 feet long at top. Irrigates 220,000 acres; cost, \$1,000,000

Strawberry Valley, Utah, in the heart of the lofty Wasatch Range, has been converted into a large lake by means of a dam in Strawberry River. The stored waters are turned through a tunnel four miles long, piercing the range, and dropped into Utah Valley on the Western slope. By means of sixty miles of cement-lined canals skirting the mountains, an area of 60,000 acres of excellent agricultural land has been reclaimed. The downward rush of water has been harnessed, and the surplus power developed is leased to several of the towns in the valley.

Reclamation Past and Future

Fifteen years have passed since reclamation became a Federal policy,—a short period in a nation's life if measured only in time, but one of historic importance when measured by achievement and progress. In this brief span of years the Service has completed sixteen notable structures of masonry and concrete, controlling the floods of torrential streams, has excavated more than 10,000 miles of canals, many of which carry whole rivers, and seventy miles of tunnels, mostly in mountains. It has to its credit the highest dam, the longest tunnel and the largest storage reservoir for irrigation in the world.

By an investment of \$120,000,000, which is repayable by the farmers in twenty years, the productive territory of the nation has been expanded by more than a million acres, and there has resulted an annual increase in its food supply valued at \$50,000,000. Where once the wilderness reigned, the hearth-stones of 200,000 people have been erected, and a citizenship established which constitutes a new bulwark of American liberty, and bulkhead the flood waters of anarchy. Thrilling, dynamic, and inspirational, this work quickens patriotic impulses and stimulates love for the republic that has promoted it.

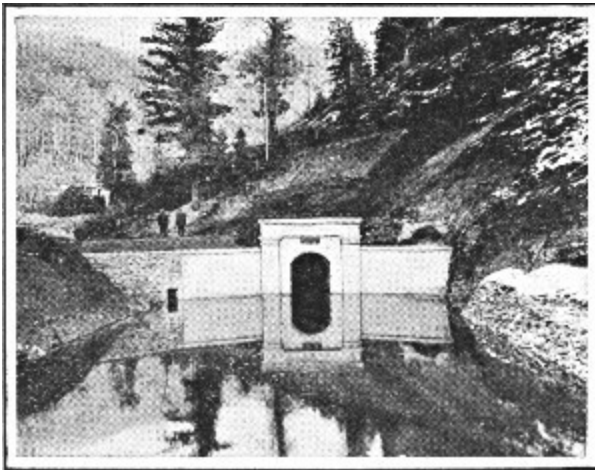


ELEPHANT BUTTE DAM, NEW MEXICO

Height, 318 feet; length at top, 1674 feet.
Creates largest artificial reservoir for irrigation in
the world

Reclamation Plans

The plans of the Reclamation Service for the present and for the immediate future are centered upon the completion of twenty-six projects, embracing a total of 3,118,000 acres. To date, engineering works have made available an adequate water supply for approximately 1,800,000 acres. In 1917, water was applied to about 1,200,000 acres, and the gross value of



OUTLET OF THE STRAWBERRY VALLEY
TUNNEL

Four miles long, and connecting Pacific Ocean
drainage with that of our great interior basin

crops was nearly \$60,000,000. It is conservative to state that with the irrigation of all the lands included in these projects, 50,000 families will have been established on individual farms. The taxable wealth will be augmented by \$300,000,000, and our annual returns from crops increased by \$100,000,000. The area included in these projects exceeds the total cropped acreage of Connecticut, Vermont, Massachusetts, Rhode Island and Delaware, and the estimated value of crops after reclamation is \$20,000,000 greater than the total returns per annum of

these states. In these estimates no account is taken of the important increment to our national wealth in the resulting growth of cities and towns, in the building of railroads, and in manufacturing and commercial institutions, which in aggregate usually equals that of land values.

In fulfilling its manifest destiny, the West has undertaken the greatest and most splendid enterprise in the world—the upbuilding of Man. It is doing for the world what the world needs most. It is producing a great people. The sentiment of it has been expressed in eloquent terms by the late Henry Grady, “A citizen,” he said, “standing in the doorway of his home, contented on his own threshold, his family gathered about him, while the evening of a well-spent day closes in scenes that are dearest,—he shall save the republic when the drum tap is futile and the



THE NEW HOME ON THE DESERT

when the drum tap is futile and the

barracks are deserted.” This pregnant sentence epitomizes the final chapter of our Romance of the Desert, as it is now being written in the arid West.



PRODUCTS OF THE DESERT

SUPPLEMENTARY READING

RECLAIMING THE ARID WEST *By George Wharton James*

UNCLE SAM'S OUTDOOR MAGIC *By P. K. Fitzhugh*

DOCUMENTS ISSUED BY THE U. S. RECLAMATION SERVICE AND
DEPARTMENT OF AGRICULTURE.

* Information concerning the above books may be had on application to the Editor of The Mentor.

THE OPEN LETTER

On reading Mr. Blanchard's account of the work of reclaiming the desert for the people, the natural question that arises in the reader's mind is, "How does one get a government farm and what are the expenses involved?" Anticipating this question, I obtained information from Mr. Blanchard in reply to it. Getting a farm is comparatively a simple proposition. Making good on it is quite another matter. In the first instance, the Government has made the way easy and inexpensive. Any citizen of the United States who has not used his homestead right is qualified to make a filing on any surveyed public lands not withdrawn from entry.



The procedure is about as follows. After a personal inspection of the vacant land desired for a home, the homeseeker makes application to the proper local Land Office or Land Commissioner and deposits filing fees of \$8 for an 80-acre or \$16 for a 160-acre tract. If the entry is made on an irrigation project, usually he must pay, in addition to the above, five per cent of the building charge, and, when due, the annual charge for operation and maintenance. On the projects containing lands now open to entry, the building charge ranges from \$30 to \$75 per acre, and is payable in twenty years without interest. The charge for operation and maintenance averages about \$1.50 per acre, and is payable annually. This charge will vary on the different projects and according to the amount of water used. Summing up the initial cost of obtaining a Government irrigated farm of 40 acres, the settler will find it necessary to expend \$6.50 for filing fees; if the construction charge is \$50 per acre, he will pay \$100 for the first instalment and, when due, about \$60 for the operation and maintenance assessment—a total of approximately \$166.50.

In the second, third, fourth and fifth years following, the only payments required are the annual charges for operation and maintenance, about \$60. Thus in five years the settler is obliged to pay to the Government approximately \$406.50.

The balance of the unpaid construction charges are payable in fifteen annual instalments, beginning on December 1 of the fifth year, the first five of which are each 5 per cent and the remaining ten 7 per cent of the construction charge, without interest for the deferred payments.

The Secretary of the Interior is authorized to require the reclamation for agricultural purposes and the cultivation of one-fourth of the irrigable area within three full irrigation seasons, and of one-half of the irrigable area within five full seasons. Actual residence, covering a period of three years, is obligatory.

On the basis of a construction charge of \$50 per acre, and an annual operation and maintenance charge of \$1.50 per acre, the entire investment of the settler at the end of the twenty years' term will be as follows:

Filing fees	\$6.50
Construction charges	2,000.00
Operation assessments	1,700.00
Total	\$3,706.50

For a farm on public lands outside of an irrigation project the only payment required is the filing fee at the time of entry.



How much capital should a man have to take up an irrigated farm? It is extremely difficult to answer this question. Experience has shown that success depends about 75 per cent on the individual. The chief requisites for success may be stated as *energy, business ability, judgment, capital and experience*. Experience has been placed last for the reason that it has been shown repeatedly that experience in farming in humid regions is not a particularly valuable asset in irrigated farming.

It may, therefore, be said that the average man who takes up a Government irrigated farm of 40 acres should possess a capital of at least \$2,500, and not less than \$4,000 if he undertakes to subdue an 80-acre tract.



As we ride on the overland trains across the great desert stretches of the far West, our eyes are fed full of color. In the midst of the riot of rich tints, we turn eagerly to occasional green spots that relieve the blazing beauty of the landscape, and our glances linger there with a sense of rest. These green spots are desert farms—fresh oases where, under the spell of water, the soil is wakening from its centuries of slumber and yielding up its stored wealth. These desert farms make a strong, human appeal to the passing traveler. They tell an assuring story of man's return to the soil and of rich returns from the soil—a story of well-being and content attained, in substantial, comfortable homes. In some such modest way as pictured on the preceding page the settler begins his conquest of the desert. The measure of his

success and prosperity lies in his hands. The wealth is there ready to be harvested.

A handwritten signature in cursive script, reading "W. D. Moffat". The signature is written in dark ink and is positioned above the word "EDITOR".

EDITOR

W. D. Moffat
EDITOR

Uncle Sam a Generous Giver



he American Government has given of its resources as no nation ever did before or ever can again. To thirteen Western States alone over a hundred million acres have been given. The peoples of the world have been called in and tendered homes. Now, out of an area within the United States of a billion and a half acres of public domain, we have left as public lands subject to disposal as homesteads and otherwise less than 280,000,000 acres, not one-half of which, it may safely be said, will ever prove to be cultivable. In one year the Department of the Interior issued over 60,000 patents to land—donations from the nation to the courageous pioneer. Any man who finds gold, silver, copper or other minerals on his grant has them for the asking—a prize for discovery.

All the revenue from the sale of public lands (less five per cent, which goes to the States) goes into a fund for the building of irrigation works to reclaim the desert. Over a hundred million dollars have been so spent, which is, however, no more than a loan to the farmers. Before attempting the governmental construction of such work the Federal Government said to the States, "If you will irrigate the lands of your State, or if there are private individuals who will do this work, we will give you whatever land you desire up to 1,000,000 acres each, and set it apart for ten years while you try the experiment." Was there ever a more generous method taken of populating and developing a new land? Those that took its lands were not asked for even so much as the cost of their administration.

In doing all this with so lavish a hand the Government has been expressing the generous instinct of the people and their absorbing determination to "go forth and find." For a hundred years and more this quest has been the drama of our national life. It has given color to

our civilization and buoyancy to the hearts of the people. It has been a century of revelation, and as yet we have only the most superficial knowledge of what this land is, of what it will yield to research, and how it may be best used.

But in all our giving we have been guided by a purpose—the land that we gave was to be converted from wilderness into homes, or from rock into metal. We gave to the homesteader, with a condition—*the land was to be used*. We gave our swamp lands, but to be reclaimed. We found our coal lands going as farms and we put a price upon them. We saw our forests being swept clean or monopolized and we held them out for the mass. Use! Use by as many as possible! The superior use! These were the things we wished and these gave form to our legislation. The homesteader may have 320 acres if it is dry farming or grazing land. But he cannot have it as a speculation. It must be made a home and brought into the body of the world's producing area by cultivation. The Government was generous, but it had no intention of being a spendthrift.

Lands and resources are at the full service of the people. And yet the romantic enterprise of revealing America is not done. To get from our resources their fullest use—this is our goal. And this is nothing less than a challenge to the capacity of democracy.

Abstract from Report of the Secretary of the Interior.

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MAKE THE SPARE
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RECLAIMING THE DESERT, VOL. 6, NUM. 17, SERIAL NO. 165,
OCTOBER 15, 1918 ***

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